EVERY VERTEX OF A MINIMAL TUTTE OBSTRUCTION HAS NEIGHBOURS IN AT LEAST THREE COMPONENTS OF THE GRAPH MINUS THAT SET

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Editorial note. All delivered mathematical text is the author's own, copied from the pinned author source sources/combinatorics-fl04-vspace-1-5cm-perfect-matching-in-p/pinned-original-source.tex; no statement or proof step was rewritten, reordered or removed.

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Editorial note. The proof cites nothing and needs no navigation stub.

Allowed standard prerequisites (editorial).

- The Tutte-violation relation for a set U and the notion of a minimal set satisfying it. Stated by the author immediately before the counted lemma (the display labelled eq:tutte-violation at source line 660, delivered verbatim as a context span of this card). Fixes the property that the minimal set U has more components in G_p - U than its own size, which the proof uses to count components after deleting a vertex.

Lemma 0.1. *Let U be a minimal set of vertices satisfying (1). Then every vertex $v \in U$ has neighbours (in G) in at least three components of $G_p - U$.*

$$\operatorname{odd}(G_p - U) \geq |U| + 2 \tag{1}$$

Proof. Suppose towards contradiction that there is a minimal set U of size u satisfying (1), with $\ell > u$ components in $G_p - U$ and containing a vertex v whose neighbourhood in G intersects at most two components of $G_p - U$.

Consider $U' = U \setminus \{v\}$. Clearly, $|U'| = u - 1$. Observe that since v is connected to at most two components in $G_p - U$, there are at least $\ell - 1 > u - 1 = |U'|$ components in $G_p - U'$. But then U' is an obstruction, contradicting the minimality of U. □

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